

Steven M. Wise

Curriculum Vitae



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Current Appointment

Mathematics Department, University of Tennessee

Professor, August 2017 – present

Associate Head and Director of Graduate Studies, April 2025 – present

Interim Department Head, January 2020

Associate Professor, August 2012 – July 2017

Assistant Professor, August 2007 – July 2012

Broad Research Impact

About 130 publications: textbooks, journal articles, and technical reports.

Web of Science Highly Cited Researcher: Mathematics, 2020, 2022

<https://recognition.webofscience.com/awards/highly-cited/2020/>

<https://recognition.webofscience.com/awards/highly-cited/2022/>

Google Scholar Citation Metrics: over 11100 citations, h-index 55, as of May, 2026

Interdisciplinary Research Interests

Computational Mathematics and Numerical Analysis: Approximation methods for nonlinear parabolic/hyperbolic PDE; convex-concave decomposition methods for bistable gradient equations. Multilevel pre-conditioners. Fast nonlinear solvers. Efficient adaptive multigrid methods for interface problems; methods for moving boundary problems, especially phase-field and diffuse interface methods. Rigorous stability and convergence of finite difference, finite element, pseudo-spectral and discontinuous galerkin methods for nonlinear and nonlocal, elliptic and parabolic PDE.

Optimization: Preconditioned Nesterov and multilevel methods for constrained and unconstrained optimization problems, especially those related to the discretization of nonlinear PDE and Calculus of Variations problems.

Mathematical Biology: Modeling tissue growth and remodeling – including cancerous tumor

growth – angiogenesis, and cancer treatment; spatiotemporal pattern formation in biological systems, morphogenesis. Deformation, decomposition, and motion of biological membranes in viscous flows.

Computational Materials Science: Growth and coarsening dynamics of heteroepitaxial thin films, long-time and large-scale coarsening of material microstructures, quantum dot formation, phase field crystal models, nonlocal models of phase transformation, crystal growth with strong interfacial anisotropy, elastically mediated phase transformations. Soft materials.

Computational Fluid Dynamics: Phase field methods for two-phase flows with large density difference and heat transport. Two-phase flows in porous media. Microstructural evolution of multiphase polymer systems, especially for organic photovoltaic (OPV) devices. Transport of ions in solution. Multi-species gas dynamics and the multi-species Boltzmann equations.

Nonlinear PDE: Existence, uniqueness, and regularity of solutions for nonlinear parabolic-hyperbolic PDE and related integro-partial differential equations (IPDE), especially for models related to phenomena described above.

Nonlinear Analysis: Γ convergence and asymptotic analyses for diffuse interface and diffuse domain problems.

University Education

Ph.D., University of Virginia, Engineering Physics; May 2003.

Advisor: W.C. Johnson

Dissertation: “Diffuse Interface Model for Microstructural Evolution of Stressed, Binary Thin Films on Patterned Substrates.”

Graduate Studies, Virginia Tech and Penn State, Mathematics; August 1996–December 1998.

Coursework: Analysis (Real, Complex, Functional, and Numerical), Applied Mathematics.

B.S., Clarion University of Pennsylvania, Mathematics; May 1996.

Minor (Coursework Equivalent): Physics

Current External Funding

NSF Grant, DMS-2309547: PI, *Collaborative Research: Accurate and Structure- Preserving Numerical Schemes for Variable Temperature Phase Field Models and Efficient Solvers*, \$213,743, 8/23 – 7/26.

ORNL-DOE-UT-Battelle CW24420: PI, *BGK Kinetic Equations*, \$606,000, 8/24 – 7/27.
ORNL Technical Point of Contact: Daniel A. Reasor, reasorda@ornl.gov.

Previous Funding, Support, and Awards

NSF Grant, DMS-2012634: PI, *Collaborative Research: Efficient, Accurate, and Structure- Preserving Numerical Methods for Phase Field-Type Models with Applications*, \$200,000, 8/20 – 7/24.

ORNL-DOE-UT-Battelle CW24420: PI, *BGK Kinetic Equations*, \$250,200, 1/22 – 7/24.
ORNL Technical Point of Contact: Ryan Glasby, glasbyrs@ornl.gov.

LANL-ORNL-DOE-UT-Battelle RFP584197: co-PI, *Developing a better understanding of adaptive velocity grids for kinetic equations*, approx. \$60,000, 2/20 – 5/21. DOE-LANL Technical Point of Contact: Jeff Haack. (Originally funded at about \$640K, before budget cuts.)

ORNL-DOE-UT-Battelle 3400267345: PI, *Quadratic Unconstrained Binary Optimization and Regularized Problems Expressed via Steady States of Dynamical Systems*, \$40,506, 6/18 – 8/18. ORNL Technical Point of Contact: Ryan Bennink.

ORNL-DOE-UT-Battelle 3400253096: PI, *Physics-Based, Self-Consistent Simulation of Neutron Reflectivity for Thin Films of Block Copolymers with Incorporated Nanoparticles*, \$138,834, 7/17 – 9/18. ORNL Technical Points of Contact: Rajeev Kumar, Valeria Lauter, Bobby Sumpter.

Dresden Fellowship: Technische Universität Dresden, Germany, € 10,500, 9/1/17 – 11/30/17.

ORNL-DOE-UT-Battelle 3400252731: PI, *Modeling and Simulation of Ion Transport in Polymer Systems and Novel Discrete Ordinate Methods for Modeling Radiation Transport via Kinetic Equations*, \$60,285, 6/17 – 5/18. ORNL Technical Point of Contact: Cory Hauck.

IMA Conference Grant: co-PI, *The 2017 Barrett Lectures: the Mathematical Foundations of Data Science*, \$5,000, 4/17 – 3/18.

NSF Grant, DMS-1719854: PI, *Efficient, Adaptive, and Convergent Numerical Methods for Phase Field Equations with Applications*, \$200,000, 8/17 – 11/21.

NSF Grant, DMS-1700494: co-PI, *The 2017 Barrett Lectures: the Mathematical Foundations of Data Science*, \$16,000, 4/17 – 3/18.

NSF Grant, DMS-1418692: PI, *Efficient, Adaptive, and Convergent Numerical Methods for Phase Field and Phase Field Crystal Equations with Applications*, \$105,000, 9/14 – 8/18.

NSF Grant, DMS-1338314: co-PI, *Southeastern-Atlantic Regional Conference on Differential Equations 2013*, \$21,938.00, 8/13 – 4/14.

NSF Grant, DMS-1115390: Lead PI, *Collaborative Research: Stable and Efficient Convexity-Splitting Schemes for Bistable Gradient PDEs*, \$160,000, 7/11 – 6/14.

NSF Grant, DMS-0818030: PI, *Collaborative Research: Multiscale Modeling of Solid Tumor Growth*, \$70,000, 9/08 – 8/11.

Workshop Support: Lead Organizer, *Investigative Workshop: Solid Tumor Modeling*, National Institute for Mathematical and Biological Synthesis (NIMBioS), approx. 40 participants, 1/19/11 – 1/21/11.

Excellence in Research and Creative Achievement Award: University of Tennessee, Spring 2025, \$3000.

Senior Research and Creative Achievement Award: College of Arts and Sciences, Natural Sciences Category, University of Tennessee, Fall/Winter Convocation 2022.

Mid-Career Faculty Research Achievement Award: College of Arts and Sciences, University of Tennessee, Fall/Winter Convocation 2014, \$500.

Junior Faculty Research Achievement Award: College of Arts and Sciences, University of Tennessee, Fall/Winter Convocation 2010, \$1000.

Best Graduate Teaching Award: Mathematics Department, University of Tennessee, 2012-13,

and 2013-14.

Best Graduate Advising Award: Mathematics Department, University of Tennessee, 2016-17.

Prior and Visiting Appointments

Dresden Senior Fellow, Technische Universität Dresden, September 2017 – November 2017.

Visiting Associate Researcher, University of California, Irvine, Mathematics Department, March 2015 – June 2015.

Visiting Assistant Professor, University of California, Irvine, Mathematics Department, July 2006 – July 2007.

Post-Doctoral Researcher, University of California, Irvine, Mathematics Department, July 2004 – June 2006.

Advisor: John S. Lowengrub

Post-Doctoral Researcher, University of California, Irvine, Biomedical Engineering and Mathematics Departments, March 2003 – June 2004.

Advisors: John S. Lowengrub and Vittorio Cristini

Research Assistant, University of Virginia, Engineering Physics Program, August 1999 – March 2003.

Scientist, United States Navy, NSWC-Dahlgren (Dahlgren, Virginia), January 1999 – August 1999.

Teaching Assistant–Primary Instructor, Penn State, Mathematics Department, August 1998 – December 1998.

Research Assistant, Virginia Tech, Mathematics Department, May 1997 – August 1998.

Teaching Assistant, Virginia Tech, Mathematics Department, August 1996 – April 1997.

Textbooks and Research Monographs

1. A.J. Salgado and **S.M. Wise**, *Classical Numerical Analysis: A Comprehensive Course*, Cambridge University Press (2023) 937 pages.
<https://www.cambridge.org/9781108837705>
<https://doi.org/10.1017/9781108942607>
2. A.J. Salgado, **S.M. Wise**, and Calvin Wong, *Multigrid Methods: Axiomatic Convergence Theory for Linear and Weakly Nonlinear Problems*, De Gruyter (2026) approx. 500 pages.
<https://doi.org/10.1515/9783111354880>
3. T. Luong, T. Mengesha and **S.M. Wise**, *Asymptotic Analysis and Perturbation Theory: Matched Asymptotic, Multiple Scale, and Homogenization Methods*, (in preparation, contract pending, 70% complete), approx. 400 pages.
4. **S.M. Wise**, *Non-Equilibrium Thermodynamics*, (in preparation, 20% complete), approx. 200 pages.

Journal Publications Since 2022

1. W. Chen, J. Jing, C. Wang, X. Wang and **S.M. Wise**, *A Modified Crank-Nicolson Numerical Scheme for the Flory-Huggins Cahn-Hilliard Model*, Commun. Comput. Phys. 31 (2022), 60–93.
<https://doi.org/10.4208/cicp.0A-2021-0074>
2. C. Liu, C. Wang, **S.M. Wise**, X. Yue, and S. Zhou, *An Iterative Solver for the Poisson-Nernst-Planck System and its Convergence Analysis*, J. Comput. Appl. Math. 406 (2022) 114017 (13 pages).
<https://doi.org/10.1016/j.cam.2021.114017>
3. C. Liu, C. Wang, Y. Wang, and **S.M. Wise**, *Convergence Analysis of the Variational Operator Splitting Scheme for a Reaction-Diffusion System with Detailed Balance*, SINUM (SIAM J. Numer. Anal.) 60 (2022) 781–803.
<https://doi.org/10.1137/21M1421283>
4. K. Cheng, C. Wang, **S.M. Wise**, and Y. Wu, *A Third Order Accurate in Time, BDF-Type Energy Stable Scheme for the Cahn-Hilliard Equation*, Numer. Math. Theor. Meth. Appl. 15 (2022) 279–303.
<https://doi.org/10.4208/nmtma.0A-2021-0165>
5. L. Dong, C. Wang, **S.M. Wise**, and Z. Zhang, *Optimal Rate Convergence Analysis of a Numerical Scheme for the Ternary Cahn-Hilliard System with a Flory-Huggins-deGennes Energy Potential*, J. Comp. Appl. Math. 415 (2022) 114474 (18 pages).
<https://doi.org/10.1016/j.cam.2022.114474>
6. E. Habbershaw and **S.M. Wise**, *A Progress Report on Numerical Methods for BGK-Type Kinetic Equations*, TRACE: Faculty Publications and Other Works, Mathematics, The University of Tennessee (2022) Report number 10.
https://trace.tennessee.edu/utk_mathpubs/10/
7. M. Yuan, W. Chen, C. Wang, **S.M. Wise**, and Z. Zhang, *A Second Order Accurate in Time, Energy Stable Finite Element Scheme for the Flory-Huggins-Cahn-Hilliard Equation*, Adv. Appl. Math. Mech. 14 (2022) 1477–1508.
<https://doi.org/10.4208/aamm.0A-2021-0331>
8. M. Punke, **S.M. Wise**, A. Voigt, and M. Salvalaglio, *Explicit Temperature Coupling in Phase-Field Crystal Models of Solidification*, Modelling Simul. Mater. Sci. Eng. 30 (2022) 074004 (18 pages).
<https://doi.org/10.1088/1361-651X/ac8abd>
9. C. Wang and **S.M. Wise**, *A Thermodynamically-Consistent Phase Field Crystal Model of Solidification with Heat Flux*, J. Math. Study 55 (2022) 1–21.
<https://doi.org/10.4208/jms.v55n4.22.01>
10. X. Chen, C. Wang, and **S.M. Wise**, *A Preconditioned Steepest Descent Solver for the Cahn-Hilliard Equation with Variable Mobility*, Int. J. Numer. Anal. Model. 19 (2022) 839–863.
<http://www.math.ualberta.ca/ijnam/Volume-19-2022/No-6-22/2022-06-06.pdf>
11. J.H. Park, A.J. Salgado, and **S.M. Wise**, *Benchmark Computations of the Phase Field Crystal and Functionalized Cahn-Hilliard Equations via Fully Implicit, Nesterov Accelerated Schemes*, Commun. Comput. Phys. 33 (2023) 367–398.
<https://doi.org/10.4208/cicp.0A-2022-0117>
12. K. Cheng, C. Wang, and **S.M. Wise**, *An Energy Stable Finite Difference Scheme for the*

- Ericksen-Leslie System with Penalty Function and its Optimal Rate Convergence Analysis*, Commun. Math. Sci. 21 (2023) 1135–1169.
<https://doi.org/10.4310/CMS.2023.v21.n4.a10>
13. M. Punke, **S.M. Wise**, A. Voigt, and M. Salvalaglio, *Improved Time Integration for Phase-Field Crystal Models of Solidification*, Proc. Appl. Math. Mech. 23 (2023) e202200112 (6 pages).
<https://doi.org/10.1002/pamm.202200112>
 14. K. Cheng, C. Wang, and **S.M. Wise**, *High Order Accurate and Convergent Numerical Scheme for the Strongly Anisotropic Cahn-Hilliard Model*, Numer. Methods Partial Diff. Eq. 39 (2023) 4007–4039.
<https://doi.org/10.1002/num.23034>
 15. X. Tang, S. Li, J.S. Lowengrub, and **S.M. Wise**, *Phase Field Modeling and Computation of Vesicle Growth or Shrinkage*, J. Math. Biol. 86 (2023) 97 (31 pages).
<https://doi.org/10.1007/s00285-023-01928-2>
 16. C. Liu, C. Wang, **S.M. Wise**, X. Yue, and S. Zhou, *A Second Order Accurate, Positivity Preserving Numerical Method for the Poisson-Nernst-Planck System and Its Convergence Analysis*, J. Sci. Comput. 97 (2023) 23 (35 pages).
<https://doi.org/10.1007/s10915-023-02345-9>
 17. Shibin Dai, Joseph Renzi, and **Steven M. Wise**, *Gamma Convergence of the DeGennes-Cahn-Hilliard Energy*, Comm. Math. Sci. 21 (2023) 2131–2144.
<https://doi.org/10.4310/CMS.2023.v21.n8.a3>
 18. Y. Guo, C. Wang, **S.M. Wise**, Z. Zhang, *Convergence Analysis of a Positivity-Preserving Numerical Scheme for the Cahn-Hilliard-Stokes System with Flory-Huggins Energy Potential*, Math. Comp. 93 (2024) 2185–2214.
<https://doi.org/10.1090/mcom/3916>
 19. C. Wang, J. Wang, **S.M. Wise**, Z. Xia, and L. Xu, *Convergence Analysis of a Temporally Second-Order Accurate Finite Element scheme for the Cahn-Hilliard-Magnetohydrodynamics System of Equations*, J. Comput. Appl. Math. 436 (2024) 115409 (17 pages).
<https://doi.org/10.1016/j.cam.2023.115409>
 20. E. Habbershaw and **S.M. Wise**, *Year-2 Progress Report on Numerical Methods for BGK-Type Kinetic Equations*, TRACE: Faculty Publications and Other Works, Mathematics, The University of Tennessee (2024) Report number 11.
https://trace.tennessee.edu/utk_mathpubs/11/
 21. Y. Sun, J. Wu, M. Jiang, **S.M. Wise**, and Z. Guo, *A Thermodynamically Consistent Phase-Field Model and an Entropy Stable Numerical Method for Simulating Two-Phase Flows with Thermocapillary Effects*, Appl. Numer. Math. 206 (2024) 161–189.
<https://doi.org/10.1016/j.apnum.2024.08.010>
 22. M. Punke, M. Salvalaglio, and A. Voigt, and **S.M. Wise**, *A Non-Isothermal Phase-Field Crystal Model with Lattice Expansion: Analysis and Benchmarks*, Modelling Simul. Mater. Sci. Eng. 33 (2025) 025007 (16 pages).
<https://doi.org/10.1088/1361-651X/ada784>
 23. E. Habbershaw, R.S. Glasby, J.R. Haack, C.D. Hauck, and **S.M. Wise**, *Asymptotic Relaxation of Moment Equations for a Multi-species, Homogeneous BGK Model*, SIAP (SIAM

- J. Appl. Math.) 85 (2025) 294–313.
<https://doi.org/10.1137/23M1596314>
24. T. Luong and **S.M. Wise**, *A Nonnegative Weak Solution to the Phase Field Crystal Model with Degenerate Mobility*, Discrete Cont. Dyn. Syst. Series B 30 (2025) 3288–3309.
<https://doi.org/10.3934/dcdsb.2025023>
 25. A. Christlieb, K. Promislow, Z. Tan, S. Wang, B. Wetton, **S.M. Wise**, *Benchmark Computation of Morphological Complexity in the Functionalized Cahn-Hilliard Gradient Flow*, Commun. Comput. Phys. 37 (2025) 877–920.
<https://doi.org/10.4208/cicp.0A-2023-0190>
 26. E. Habbershaw, C.D. Hauck, and **S.M. Wise**, *Implicit Update of the Moment Equations for a Multi-Species, Homogeneous BGK Model*, SINUM (SIAM J. Numer. Anal.) 63 (2025) 881–907.
<https://doi.org/10.1137/24M165421X>
 27. A.E. Diegel, C. Wang, and **S.M. Wise**, *Convergence Analysis of a Preconditioned Steepest Descent Solver for the Cahn-Hilliard Equation with Logarithmic Potential*, Int. J. Numer. Anal. Model. 22 (2025) 483–509.
<https://doi.org/10.4208/ijnam2025-1021>
 28. T. Luong, T. Mengesha, **S.M. Wise**, and M.H. Wong, *A Diffuse Domain Approximation with Transmission-Type Boundary Conditions I: Asymptotic Analysis and Numerics*, Int. J. Numer. Anal. Model. 22 (2025) 694–727.
<https://doi.org/10.4208/ijnam2025-1030>
 29. T. Luong, T. Mengesha, **S.M. Wise**, and M.H. Wong, *A Diffuse Domain Approximation with Transmission-Type Boundary Conditions II: Gamma Convergence*, Int. J. Numer. Anal. Model. 22 (2025) 728–744.
<https://doi.org/10.4208/ijnam2025-1031>
 30. Y. Guo, C. Wang, **S.M. Wise**, and Z. Zhang, *A Uniquely Solvable and Positivity-Preserving Finite Difference Scheme for the Flory-Huggins-Cahn-Hilliard Equation with Dynamical Boundary Condition*, J. Comput. Appl. Math. 472 (2026) 116789 (21 pages).
<https://doi.org/10.1016/j.cam.2025.116789>
 31. S. Tang, **S.M. Wise**, J.S. Lowengrub, and Z. Guo, *Membrane Heterogeneity Driven Dynamics of Multicomponent Vesicles in Shear Flow*, J. Fluid. Mech. 1030 (2026) A47 (36 pages).
<https://doi.org/10.1017/jfm.2026.11276>
 32. J. Meng, S. Tang, **S.M. Wise**, and Z. Guo, *A GPU-Accelerated Matrix-Free FAS Multigrid Solver for Navier-Stokes Equations with Memory-Efficient Implementations* (in review).
<https://doi.org/10.48550/arXiv.2510.11152>
 33. J.H. Park, A.J. Salgado, and **S.M. Wise**, *A Perturbed Preconditioned Gradient Descent Method for the Unconstrained Minimization of Composite Objectives*, J. Sci. Comput. (in review).
<https://doi.org/10.48550/arXiv.2512.19532>
 34. Z. Zhang, S. Li, J.S. Lowengrub, and **S.M. Wise**, *An Efficient Constant-Coefficient MSAV Scheme for Computing Vesicle Growth and Shrinkage*, Commun. Comput. Phys. (in review).
<https://doi.org/10.48550/arXiv.2601.10057>

Graduate Students

Former (alphabetical by Last Name)

1. Andreas Aristotelous, Math UTK, PhD 2011. Thesis: "Adaptive Discontinuous Galerkin Finite Element Methods for a Diffuse Interface Model of Biological Growth." (O. Karakashian, UTK, was the primary advisor.) First position: postdoc at Duke University and SAMSI.
2. Kylie Berry, Math UTK, PhD 2019. Topic: "Energy-Based Blending Models in Peridynamics." (T. Mengesha, UTK, was the co-advisor, and P. Seleson was the technical point of contact at ORNL for this project.) First position: Assistant Professor of Mathematics at Dalton State College, Dalton, GA.
3. Craig Collins, Math UTK, PhD 2015. Thesis: "Domain Decomposition Methods for Discontinuous Galerkin Approximations of Elliptic Problems." (O. Karakashian, UTK, was the primary advisor.) First position: Assistant Professor of Mathematics at Murray State University, Murray, KY.
4. John Cummings, Math UTK, PhD 2017. Topic: "Simulating Block Polymer-Solvent Mixtures for Organic-Photovoltaic Devices." (R. Kumar was the technical point of contact at ORNL for this project.) First position: Visiting assistant professor, Berry College, Rome, GA.
5. Amanda Diegel, Math UTK, PhD May 2015. Thesis: "Numerical Analysis of Convex Splitting Schemes for Cahn-Hilliard and Coupled Cahn-Hilliard-Fluid-Flow Equations." First position: Postdoctoral researcher, Mathematics Department, Louisiana State University, Baton Rouge, LA.
6. Wenqiang Feng Math UTK, PhD 2017. Topic: "Linearly Preconditioned Nonlinear Solvers for Phase Field Equations Involving p-Laplacian Terms." First position: Data scientist, DST Systems Inc, Kansas City, MO.
7. Stephen Galloway, Math UTK, MS 2017. Project: "Numerical Simulation of Solidification with Strong Interface Anisotropy." Continued in PhD program at UTK.
8. Stephen Galloway, Math UTK, PhD 2020. Thesis: "Graphene Formation using the PFC Model with a Vapor Phase." First position: Assistant Professor, Mathematics Department, St. Andrews University, Laurinburg, NC.
9. Zhen Guan, Math UTK, PhD August 2012. Thesis: "Numerical Analysis of First and Second Order Unconditionally Energy Stable Schemes for Nonlocal Cahn-Hilliard and Allen-Cahn Equations." First position: Visiting Assistant Professor, Mathematics Department, University of California, Irvine, CA.
10. Evan Habbershaw, Math UTK, PhD 2025. Project: "Numerical Methods for Multi-Species BGK Models." (C. Hauck was the technical point of contact at ORNL for this project.) First position: Visiting Assistant Professor, Mathematics Department, Pennsylvania State University, State College, PA.
11. Ben Plumridge, Math UTK, PhD 2025. Project: "Neural Network-Based Adaptive Filtering of the Spherical Harmonic Method in Radiation Transport." (Cory Hauck, ORNL, was the primary advisor.)
12. Kai Kang, Math UTK, MS May 2012. Project: "A Second-Order Accurate Energy Stable Scheme for the Cahn-Hilliard-Hele-Shaw System of Equations: Application to

Tumor Growth.” Went on to complete PhD at UTK, 2017.

13. Jea-Hyun Park, Math UTK, PhD 2021. Thesis: “Preconditioned Nesterov’s Accelerated Gradient Descent Method and its Applications for Nonlinear PDE.” (A. Salgado, UTK, was the co-advisor.) First position: Visiting Assistant Professor, Mathematics Department, University of California, Santa Barbara, CA.

Current (alphabetical by last name)

1. Charlotte Beckford, Math UTK, PhD 2026 (expected). Project: “Thermodynamically Consistent Models of Polymer Batteries.” (R. Kumar is the technical point of contact at ORNL for this project.) My UTK advising percentage: 100%.
2. Sam Gruber, Math UTK, PhD 2027 (expected). Project: “Phase Transformations in SrCoOx Materials for Neuromorphic Computing Architectures.” (V. Starchenko and P. Ganesh are the technical points of contact at ORNL for this project.) My UTK advising percentage: 100%.
3. Jack Haight, Math UTK, PhD 2027 (expected). Project: “Numerical Methods for Kinetic Equations.” (C. Hauck is the technical point of contact at ORNL for this project.) My UTK advising percentage: 100%.
4. Suleman Shahid, Math UTK, PhD 2027 (expected). Project: TBD. (Abner Salgado, UTK, is the co-advisor.) My UTK advising percentage: 50%.
5. Calvin Wong, Math UTK, PhD 2026 (expected). Project: “Numerical Methods for and Asymptotic Analyses of Diffuse Domain Problems.” (C. Hauck is the technical point of contact at ORNL for a companion project.) My UTK advising percentage: 100%.

Postdoctoral Researchers

Former

Toai Luong, 2022-2025, PhD from University of Alabama. First position after UTK: Visiting Assistant Professor, Virginia Commonwealth University, Richmond, VA.

References

1. **Wenbin Chen**, Professor
School of Mathematics
Fudan University
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2. **Qiang Du**, Fu Foundation Professor of
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3. **John S. Lowengrub**, Chancellor's
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